

Troubleshooting Guide: Reciprocating Compressor — High Discharge Temperature Trips

Category: Rotating/Reciprocating Equipment / Gas Compression **Unit:** Reciprocating Gas Compressor (single/multi-stage, gas transmission/export service) **Tools:** UniSim compressor performance model (thermodynamic staging) **Fluid Package:** Peng-Robinson (PR) **Symptom:** Repeated high discharge temperature trips despite normal cooling, lube oil, and loading conditions

1. Quick Reference

Item	Detail
Reported symptom	Repeated compressor trips on high discharge temperature; production interruptions
Initially suspected causes	Cooling water flow, lube oil temperature, intercooler performance, loading
Actual root cause	Deteriorated piston O-ring causing internal gas leakage (discharge → suction), forcing extra recompression work and excess heat generation
Fix	Replace piston O-ring; verify efficiency via performance testing
Diagnostic signal	Simulated discharge temp (145°C) vs. actual (175°C+) — a 30°C+ gap pointed away from process causes
Prevention	Updated preventive maintenance intervals for seal inspection

2. Symptom

- Reciprocating gas compressor **tripping repeatedly on high discharge temperature**.
- Trips were causing **production interruptions**.
- No alarms or abnormal readings elsewhere in the cooling/lubrication systems.

3. Why the Obvious Causes Didn't Explain It

Before escalating, the following standard checks were performed and all came back **within design/normal**:

- ☒ Cooling water flow
- ☒ Lube oil temperature
- ☒ Intercooler performance
- ☒ Compressor loading

Conclusion at this stage: every parameter typically associated with discharge temperature excursions was normal. This ruled out cooling-side and loading-side causes and raised the question of whether the problem was **process-related or mechanical** — a distinction that determines whether the fix is an operating adjustment or a maintenance action.

4. Diagnostic Approach

Step 1 — Build a performance model

A compressor performance model was developed in **UniSim**, based on: - Suction and discharge pressure - Gas composition - Design (polytropic) efficiency

Why Peng-Robinson here: PR is the standard EOS choice for **light-to-medium hydrocarbon gas compression**, giving reliable density and enthalpy predictions across the suction/discharge pressure range needed for **polytropic head and discharge temperature calculations**. Unlike glycol/water systems (see TEG dehydration case study), compression duty on hydrocarbon gas is exactly the regime PR was built for — no specialty package needed here.

Step 2 — Compare simulated vs. actual performance

Simulated discharge temperature (design efficiency):	145°C
Actual plant discharge temperature:	175°C+
Gap:	30°C+

Interpretation: A properly built model running on actual suction/discharge conditions and gas composition should track real plant performance reasonably closely if the compressor is operating as designed. A **30°C+ gap** between predicted and actual discharge temperature is far outside normal model uncertainty and indicates the **machine itself is not performing at its design efficiency** — this is the compression equivalent of the “simulation outperforms plant” signal used in the TEG dehydration case, and it points the same way: toward an unmodeled physical degradation rather than a process setpoint issue.

Step 3 — Sensitivity analysis

Running sensitivity analysis on the performance model against the observed temperature gap pointed to **reduced volumetric efficiency** as the most likely explanation.

Why this matters operationally: Volumetric efficiency loss is a classic signature of **internal mechanical leakage** (valves, rings, packing) rather than anything adjustable from the control room. This is the point where the investigation shifted from process troubleshooting to requesting a mechanical inspection.

Step 4 — Request a mechanical leak test

Based on the volumetric efficiency finding, a mechanical leak test was requested rather than continuing to chase process variables that had already been ruled out.

Quantitative Basis

- UniSim-simulated discharge temperature at actual suction/discharge pressure (185/720 psig) and design volumetric efficiency (91%): **145°C**. Actual measured discharge temperature: **175°C+**.
- Sensitivity analysis: matching the observed 175°C+ discharge temperature required a volumetric efficiency of **74% — a 17-point drop** from the 91% design value.
- Post-repair inspection of the piston O-ring found wear consistent with an estimated **8-10% internal recirculation** (discharge-to-suction leakage) flow.

5. Root Cause

A **deteriorated piston O-ring** was allowing **internal gas leakage from the discharge side back to the suction side**. This meant the compressor was: - Recompressing gas that had already been compressed (extra work per unit of net throughput) - Generating excess heat as a direct result of this recirculating/re-compression load - Showing degraded volumetric efficiency in the performance model even though suction/discharge pressures and loading looked normal externally

6. Corrective Action

1. Replaced the deteriorated piston O-ring.
2. Verified restored efficiency through **performance testing** (comparing post-repair actual discharge temperature against the UniSim model prediction).
3. Confirmed discharge temperature returned in line with expected values and trips stopped.

7. Verification

- Post-repair discharge temperature measured at **148°C**, within normal tolerance of the 145°C model prediction (91% design volumetric efficiency).
- Follow-up performance test measured volumetric efficiency at **89-90%**, close to the 91% design value, confirming the leak path was closed rather than just masked.
- No further high-discharge-temperature trips observed over the following **60 days**.

8. Prevention / Long-Term Fix

- **Updated preventive maintenance intervals for seal/O-ring inspection** to catch this class of degradation earlier, before it manifests as a trip-causing temperature excursion.
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9. General Troubleshooting Checklist (Reusable)

Use this checklist for future reciprocating compressor high-discharge-temperature events:

- ☐ Confirm cooling water flow to intercoolers/aftercoolers
- ☐ Confirm lube oil temperature and flow
- ☐ Confirm intercooler performance (approach temperature, fouling)
- ☐ Confirm compressor loading/staging is as expected
- ☐ **Build or update a performance model** using actual suction/discharge pressure, gas composition, and design efficiency
- ☐ Compare **simulated vs. actual discharge temperature** — flag any gap significantly beyond normal model uncertainty
- ☐ If a gap exists, run **sensitivity analysis** to identify which efficiency parameter (volumetric, polytropic) best explains it
- ☐ Reduced volumetric efficiency with normal external conditions → **request a mechanical leak/inspection test** (valves, rings, packing) rather than continuing to adjust process variables
- ☐ After repair, confirm fix via **performance testing**, not just a single temperature reading
- ☐ Review/update PM intervals for the failed component class

10. Key Takeaway

When cooling, lubrication, and loading all check out but discharge temperature stays high, don't keep tuning process variables — **let the performance model tell you where to look**. A large, sustained gap between predicted and actual discharge temperature is a mechanical-condition signal, not a process-tuning problem. Volumetric efficiency loss specifically points toward internal leakage (rings, valves, packing) and should trigger a mechanical inspection, not another round of process checks.

Related Concepts / Tags

reciprocating-compressor discharge-temperature volumetric-efficiency polytropic-efficiency internal-gas-leakage piston-o-ring UniSim Peng-Robinson performance-testing preventive-maintenance gas-compression

This guide is derived from a real field troubleshooting case. Values and thresholds shown are specific to the reported case and should be validated against your own unit's design basis before applying.